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Convergence of Newton-Cotes Quadrature

+12p

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Abstract

We revisit the composite trapezoidal and Simpson 1/3 rules and chart their behaviour on four representative integrals, two smooth and two with endpoint singularities. On smooth integrands we recover the textbook $O(h^2)$ and $O(h^4)$ rates; on singular integrands both rules degrade together, and the high-order advantage of Simpson's rule collapses to a small constant factor. The poster summarises the results in three columns: introduction, method, and numerical evidence.

Introduction

Numerical integration is the inner loop of most scientific computing. Closed-form antiderivatives are the exception; the rule is to discretise the integration interval and sum weighted samples of the integrand. The two simplest composite rules date to Newton and Simpson and remain in routine use as the inner kernels of adaptive quadrature libraries [1]. Despite their age, the quantitative comparison of the two rules on specific integrand classes continues to inform practical algorithm choices [2].

The fundamental question – *how many samples are needed, and where should they be placed?* – has been studied since Newton, but it continues to surface

in modern guises whenever a new class of integrand puts pressure on existing libraries.

Method

Let $f : [a, b] \rightarrow \mathbb{R}$ and partition $[a, b]$ into n equal subintervals of width $h = (b - a)/n$. The composite trapezoidal rule approximates the integral by

$$T_n(f) = \frac{h}{2} \left[f(x_0) + 2 \sum_{i=1}^{n-1} f(x_i) + f(x_n) \right]$$

with leading error $-(b-a)h^2 f''(\xi)/12$ for $\xi \in (a, b)$, provided $f \in C^2[a, b]$. The rule is second-order accurate.

With n even, Simpson's 1/3 rule groups nodes in overlapping triples and integrates the quadratic interpolant exactly:

$$S_n(f) = \frac{h}{3} \left[f(x_0) + 4 \sum_{k=1}^{n/2} f(x_{2k-1}) + 2 \sum_{k=1}^{n/2-1} f(x_{2k}) + f(x_n) \right]$$

with leading error $-(b-a)h^4 f^{(4)}(\eta)/180$ for $\eta \in (a, b)$. The rule is fourth-order accurate. Both rules were implemented in double precision IEEE 754 with Kahan-compensated summation [3].

Results

Table 1 collects absolute errors for $I_1 = \int_0^1 e^x dx$ as n doubles. The ratio column confirms the asymptotic prediction: trapezoidal errors fall by 4 times per halving of h , Simpson's by 16 times.

n	index
8	right
16	index
32	right
64	index

The convergence of absolute error is -2.00 and $\ln$ prediction

[SVG file omitted]

Log-log of Simpson's numbers

For integration changes, and rule shrinking sampled recovers

Discussion

Three textbook observations and one subtler one fall out. First, on smooth integrands Simpson's rule should be the default: the implementation cost is essentially identical to the trapezoidal rule and the asymptotic advantage is dramatic. Second, the asymptotic theory is fragile – once smoothness is lost at an endpoint, both rules degrade together. Third, adaptive routines interleave the two rules not for accuracy but for *error estimation*, exploiting Richardson extrapolation on the Simpson-minus-trapezoidal difference.

The subtler observation is that the Simpson-to-trapezoidal error ratio tends to a function of the integrand's regularity, not of h . Students who memorise $h^4 \parallel h^2$ are often surprised when, on a real singular integrand, the two rules agree to within a factor of three.

References

References

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